Dual and energy-based potentials

Overview of Large-Scale OT and Mapping Estimation

Let $\mathcal{X}$ and $\mathcal{Y}$ be complete metric spaces. Let $\mathcal{C}(\mathcal{X})$ be the space of all continuous functions on $\mathcal{X}$, and let $\mathcal{P}(\mathcal{X})$ be the space of all probability measures on $\mathcal{X}$ (respectively $\mathcal{C}(\mathcal{Y})$ and $\mathcal{P}(\mathcal{Y})$ for $\mathcal{Y}$). Let $\mu \in \mathcal{P}(\mathcal{X})$ and $\nu \in \mathcal{P}(\mathcal{Y})$ be two probability measures, and let $\Pi(\mu, \nu)$ represent the class of joint couplings $\pi \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$ such that for any measurable subsets $A \subseteq \mathcal{X}$ and $B \subseteq \mathcal{Y}$, the marginals satisfy $\pi(A \times \mathcal{Y}) = \mu(A)$ and $\pi(\mathcal{X} \times B) = \nu(B)$.

Consider a cost function $c : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^+$, and a regularization function $R : \mathcal{P}(\mathcal{X}, \mathcal{Y}) \to [0, +\infty)$.

Then the Entropic Optimal Transport problem for some $\epsilon > 0$ is defined as finding the optimal coupling $\pi^\epsilon \in \Pi(\mu, \nu)$ such that:

$$\pi^\epsilon = \arg \inf_{\pi \in \Pi(\mu,\nu)} \left\{ \mathbb{E}_{(X,Y)\sim\pi}[c(X, Y)] + \epsilon R(\pi) \right\}$$

Many regularizers have been proposed. The authors of Large Scale Optimal Transport and Mapping Estimation experimented with entropy regularization $R_e(\pi)$ and the L^2 regularization $R_{L^2}(\pi)$. Their closed form expression is biven by:

$$R(\pi) = \begin{cases} \displaystyle\int_{\mathcal{X}\times\mathcal{Y}} \left(\ln \frac{\mathrm{d}\pi(x,y)}{\mathrm{d}\mu(x)\,\mathrm{d}\nu(y)} - 1 \right) \mathrm{d}\pi(x,y), & \text{for entropy regularization,} \\ \displaystyle\int_{\mathcal{X}\times\mathcal{Y}} \left(\frac{\mathrm{d}\pi(x,y)}{\mathrm{d}\mu(x)\,\mathrm{d}\nu(y)} \right)^2 \mathrm{d}\mu(x)\,\mathrm{d}\nu(y), & \text{for } L^2 \text{ regularization.} \end{cases}$$

This primal EOT problem admits a dual representation, which is the following optimization problem:

$$(u^\epsilon, v^\epsilon) = \arg \sup_{(u,v)\in\mathcal{C}(\mathcal{X})\times\mathcal{C}(\mathcal{Y})} \mathbb{E}_{(X,Y)\sim\mu\otimes\nu}[u(X) + v(Y) + F_\epsilon(u(X), v(Y), c(X, Y))]$$

where the function F_ϵ is defined based on the regularization we use. Bellow we provide the close forms for both entropy and L^2 regularizations:

$$F_\epsilon(u(x), v(y), c(x,y)) = \begin{cases} -\epsilon \exp\left(\frac{1}{\epsilon}(u(x) + v(y) - c(x,y))\right), & \text{for entropy regularization,} \\ -\frac{1}{4\epsilon}(u(x) + v(y) - c(x,y))^2 & \text{for } L^2 \text{ regularization.} \end{cases}$$

The following algorithm computes the optimal potentials u^ϵ, v^ϵ:

Algorithm 1: Stochastic OT Computation

Input: measures μ, ν, cost function c, batch size p, learning rate γ

Discrete case: $\mu = \sum_i a_i \delta_{x_i}$, $u = u(x_i) = x_i$ finite vector (similarly for ν, v)
Continuous case: μ is a continuous measure and u is a neural network (similarly for ν, v)

while *Not Converged* **do**

Sample a batch $(x_1, \ldots, x_p)$ from μ.

Sample a batch $(y_1, \ldots, y_p)$ from ν.

$$u \leftarrow u + \gamma \sum_{i,j=1}^{p} (\nabla u(x_i) + \partial_u F_\epsilon \nabla u(x_i))$$

$$v \leftarrow v + \gamma \sum_{i,j=1}^{p} (\nabla v(x_i) + \partial_v F_\epsilon \nabla v(x_i))$$

Output: Learned potentials u, v.

In order to recover the solution π^ϵ of the regularized primal problem we use the first order optimality condition of the Frenchel-Rockafellar's duality theorem:

$$\mathrm{d}\pi^\epsilon = H_\epsilon(x,y)\mathrm{d}\mu(x)\mathrm{d}\nu(y)$$

where the function H_ϵ is derived analytically based on the regularizer:

$$H_\epsilon(x,y) = \begin{cases} \exp\left(\frac{u^\epsilon(x)}{\epsilon}\right) \exp\left(-\frac{c(x,y)}{\epsilon}\right) \exp\left(\frac{v^\epsilon(y)}{\epsilon}\right), & \text{for entropy regularization,} \\ \frac{1}{2\epsilon}(u^\epsilon(x) + v^\epsilon(y) - c(x,y)) & \text{for } L^2 \text{ regularization.} \end{cases}$$

We use the optimal transport plan π^ϵ (the solution to the EOT problem) to compute a deterministic mapping $T(x)$ via the barycentric projection. Assuming the squared

Euclidean distance is used for the barycentric cost, the projection for a given source point x is defined as:

$$T(x) = \mathbb{E}_{Y \sim \pi^\epsilon(\cdot|x)}[Y]$$

We can learn the barycentric projection using NN:

$$\theta^* = \arg\min_\theta \mathbb{E}_{X \sim \mu} \left[\mathbb{E}_{Y \sim \pi^\epsilon(\cdot|X)} \left[\|T_\theta(X) - Y\|^2 \right] \right]$$

By the law of total expectation, the nested conditional expectation simplifies to a single expectation over the joint transport plan π^ϵ:

$$\theta^* = \arg\min_\theta \mathbb{E}_{(X,Y) \sim \pi^\epsilon} \left[\|T_\theta(X) - Y\|^2 \right]$$

Using the Frenchel-Rockafellar optimality condition you defined earlier ($\mathrm{d}\pi^\epsilon = H_\epsilon(x, y)\mathrm{d}\mu(x)\mathrm{d}\nu(y)$), we rewrite this as an integral over the independent marginals:

$$\theta^* = \arg\min_\theta \mathbb{E}_{(X,Y) \sim \mu \otimes \nu} \left[\|T_\theta(X) - Y\|^2 H_\epsilon(X, Y) \right]$$

We can learn the barycentric projection using a Neural Network by minimizing the Mean Squared Error over the joint transport plan. The parameters θ are optimized as follows:

$$\theta^* = \arg\min_\theta \mathbb{E}_{X \sim \mu} \left[\mathbb{E}_{Y \sim \pi^\epsilon(\cdot|X)} \left[\|T_\theta(X) - Y\|_2^2 \right] \right]$$

Using the property $\mathrm{d}\pi^\epsilon = H_\epsilon(x, y)\mathrm{d}\mu(x)\mathrm{d}\nu(y)$, we can optimize this empirically using SGD. The explicit gradient for the quadratic cost allows us to define the following algorithm:

Algorithm 2: Optimal map learning with SGD (Quadratic Cost)

Input: input measures μ, ν; dual optimal variables u^ϵ and v^ϵ; map T_θ parameterized as a deep NN; batch size n; learning rate γ.

while *Not Converged* **do**

Sample a batch $(x_1, \ldots, x_n)$ from μ.

Sample a batch $(y_1, \ldots, y_n)$ from ν.

Update the network parameters using the empirical gradient:

$$\theta \leftarrow \theta - \gamma \sum_{i,j=1}^{n} H_\epsilon(x_i, y_j) \cdot 2(T_\theta(x_i) - y_j) \nabla_\theta T_\theta(x_i)$$

Output: Learned optimal map T_θ.

Overview of Score-Based Generative Neural Networks for Large Scale OT

Let $\mathcal{X}$ and $\mathcal{Y}$ be complete metric spaces. Let $\mu \in \mathcal{P}(\mathcal{X})$ and $\nu \in \mathcal{P}(\mathcal{Y})$ be two probability measures.

The authors of Score-based Generative Neural Networks for Large-Scale Optimal Transport generalize the notion of regularization by utilizing the class of f-Divergences. Let $f : \mathbb{R} \to \mathbb{R}$ be a strictly convex function such that $f(1) = 0$. The f-divergence between the joint transport plan π and the product measure $\mu \otimes \nu$ is defined as:

$$H_f(\pi) = D_f(\pi \mid \mu \otimes \nu) = \mathbb{E}_{(X,Y)\sim\mu\otimes\nu}\left[f\left(\frac{\mathrm{d}\pi(X,Y)}{\mathrm{d}\mu(X)\mathrm{d}\nu(Y)}\right)\right]$$

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable a-strongly convex function with convex conjugate f^* and denote $f^{*'}(u) = \partial_u f^*(u)$ for u in the domain of f.
Define the violation function:

$$V(x, y\ ;\ \phi, \psi) = \phi(x) + \psi(y) - c(x, y)$$

for $\phi \in \mathcal{C}(\mathcal{X})$, $\psi \in \mathcal{C}(\mathcal{Y})$ the dual potentials.

Then the dual objective takes the form:

$$(\phi^\epsilon, \psi^\epsilon) = \arg \min_{(\phi,\psi)\in\mathcal{C}(\mathcal{X})\times\mathcal{C}(\mathcal{Y})} \mathbb{E}_{(X,Y)\sim\mu\otimes\nu}\left[\phi(X) + \psi(Y) - H_f^*(V(x, y;\ \phi, \psi))\right]$$

where $H_f^*(u) = \epsilon f^*(\epsilon^{-1}u)$.

And the optimal solutions $(\pi^\epsilon, \phi^\epsilon, \psi^\epsilon)$ satisfy:

$$\pi^\epsilon(x, y) = M_f(V(x, y;\ \phi^\epsilon, \psi^\epsilon))\mu(x)\nu(y)$$

where $M_f(u) = f^{*'}(\epsilon^{-1}u)$.

To generate samples the authors apply Langevin Sampling:

$$\tilde{x}_t = \tilde{x}_{t-1} + \eta\nabla_x \log p(\tilde{x}_{t-1}) + \sqrt{2\eta}z_t, \quad z_t \sim N(0, \mathbf{I})$$

Similarly to the Large-Scale OT and Mapping Estimation, they approximate the optimal coupling π^ϵ via the dual potentials parameterize using NN:

$$\phi_\theta : \mathcal{X} \to \mathbb{R}, \qquad \psi_\theta : \mathcal{Y} \to \mathbb{R}$$

Bellow is the algorithm adjusted to their notation.

Algorithm 3: Density Estimation

Input: step size γ, batch size m, nets ϕ_{θ_2}, ψ_{θ_1}, datasets μ, ν, time steps $T > 0$

for $t = 1$ **to** T **do**

Sample a batch $(x_1, \ldots, x_m)$ from μ.

Sample a batch $(y_1, \ldots, y_m)$ from ν.

$$\Delta_1 \leftarrow \sum_{i,j=1}^{m} \nabla_{\theta_1}[\phi_{\theta_1}(x_i) - H^*(V(x_i, y_j \ ; \ \phi_{\theta_1}, \psi_{\theta_2}))]$$

$$\Delta_2 \leftarrow \sum_{i,j=1}^{m} \nabla_{\theta_2}[\psi_{\theta_2}(y_i) - H^*(V(x_i, y_j \ ; \ \phi_{\theta_1}, \psi_{\theta_2}))]$$

$$\theta_1 \leftarrow \theta_1 - \gamma\Delta_1$$
$$\theta_2 \leftarrow \theta_2 - \gamma\Delta_2$$

Output: Learned dual potentials ϕ_{θ_1}, ψ_{θ_2}.

After optimizing the parameters θ^* the authors proposed sampling the conditional $\pi^{\epsilon}_{Y|X=x}$ using Langevin Dynamics. Recall that the conditional optimal coupling is given by:

$$\pi^{\epsilon}_{Y|X=x}(y) = M_f(V(x, y \ ; \ \phi^{\epsilon}, \psi^{\epsilon}))\nu(y)$$

Therefore the intractable score is the following:

$$\nabla_y \log \pi^{\epsilon}_{Y|X=x}(y) = \nabla_y \log M_f(V(x, y \ ; \ \phi^{\epsilon}, \psi^{\epsilon})) + \nabla_y \log \nu(y)$$

The authors proposed using score matching techniques in order to estimate intractable score, namely:

$$s_\theta(y) \approx \nabla_y \log \nu(y)$$

For example the score network can be trained on the following loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{Y\sim\nu,\ \tilde{Y}\sim q_\sigma(\cdot|Y)} \left[\left\| s_\theta(\tilde{Y}, \sigma) - \nabla_{\tilde{Y}} \log q_\sigma(\tilde{Y} \mid Y) \right\|_2^2 \right]$$

where $q_\sigma(\tilde{Y} \mid Y)$ is a predefined noise perturbation kernel, typically a Gaussian distribution such as $N(Y, \sigma^2\mathbf{I})$.

Algorithm 4: Sampling Procedure

Input: noise levels $\tau_1 > \tau_2 < \cdots < \tau_N$, dual potentials ϕ_{θ_1}, ψ_{θ_2}, source $x \in \mathcal{X}$, time steps $T > 0$, step size $\eta > 0$

Sample $\tilde{y}_{1,0} \sim N(0, 1)$
for $\tau_i, i = 1$ **to** N **do**
  for $t = 1$ **to** T **do**
    Sample $z_i \sim N(0, \tau_i)$

$$\Delta_s \leftarrow s_\theta(\tilde{y}_{i,t-1})$$
$$\Delta_\pi \leftarrow \nabla_y \log M(V(x, \tilde{y}_{i,t-1} \; ; \; \phi_{\theta_1}, \psi_{\theta_2}))$$
$$\tilde{x}_{i,t} \leftarrow \tilde{x}_{i,t-1} + (\eta/2)(\Delta_s + \Delta_\pi) + \sqrt{\eta} z_i$$

  Initialize $\tilde{y}_{i+1,0} = \tilde{y}_{i,T}$
Output: Data samples from $\pi^{\epsilon}_{Y|X=x}$

Overview of Energy Guided Neural EOT

Let us assume that the measures μ and ν admit probability densities $\mu(x)$ and $\nu(y)$ with respect to appropriate base measures $\mathrm{d}x$ and $\mathrm{d}y$. Let any valid coupling $\pi \in \Pi(\mu, \nu)$ admit a joint density $\pi(x, y)$ with respect to $\mathrm{d}x\mathrm{d}y$.

By the disintegration theorem, the joint density can be factored into the marginal density of X and the conditional density:

$$\pi(x, y) = \pi(y \mid x)\mu(x)$$

The authors of Energy-Guided Neural Entropic Optimal Transport utilize the conditional entropy regularization:

$$R(\pi) = -\int_{\mathcal{X}} H(\pi(\cdot \mid x))\mu(x)\mathrm{d}x$$

where the Shannon entropy of the conditional distribution is defined using its density:

$$H(\pi(\cdot \mid x)) = -\int_{\mathcal{Y}} \pi(y \mid x) \log \pi(y \mid x)\mathrm{d}y$$

Substituting the conditional entropy into the regularizer, we obtain:

$$\begin{aligned} R(\pi) &= \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} \pi(y \mid x) \log \pi(y \mid x) \mathrm{d}y \right) \mu(x) \mathrm{d}x \\ &= \int_{\mathcal{X}\times\mathcal{Y}} \pi(y \mid x) \mu(x) \log \pi(y \mid x) \mathrm{d}x \mathrm{d}y \\ &= \int_{\mathcal{X}\times\mathcal{Y}} \pi(x, y) \log \frac{\pi(x, y)}{\mu(x)} \mathrm{d}x \mathrm{d}y \end{aligned}$$

To connect this to the f-divergence framework, we target the KL divergence $\mathrm{KL}(\pi \mid \mu \otimes \nu)$, which evaluates the logarithm of the ratio between the joint density and the independent product of marginals $\mu(x)\nu(y)$. By multiplying and dividing the argument of the logarithm by $\nu(y)$, we can expand the integral:

$$\begin{aligned} R(\pi) &= \int_{\mathcal{X}\times\mathcal{Y}} \pi(x, y) \log \left(\frac{\pi(x, y)}{\mu(x)\nu(y)} \nu(y) \right) \mathrm{d}x \mathrm{d}y \\ &= \int_{\mathcal{X}\times\mathcal{Y}} \pi(x, y) \log \frac{\pi(x, y)}{\mu(x)\nu(y)} \mathrm{d}x \mathrm{d}y + \int_{\mathcal{X}\times\mathcal{Y}} \pi(x, y) \log \nu(y) \mathrm{d}x \mathrm{d}y \end{aligned}$$

The first term is exactly the formulation of the KL divergence. For the second term, because $\pi \in \Pi(\mu, \nu)$, the marginalization of $\pi(x, y)$ over $\mathcal{X}$ must strictly equal the target marginal $\nu(y)$. Therefore:

$$\begin{aligned} \int_{\mathcal{X}\times\mathcal{Y}} \pi(x, y) \log \nu(y) \mathrm{d}x \mathrm{d}y &= \int_{\mathcal{Y}} \left(\int_{\mathcal{X}} \pi(x, y) \mathrm{d}x \right) \log \nu(y) \mathrm{d}y \\ &= \int_{\mathcal{Y}} \nu(y) \log \nu(y) \mathrm{d}y \\ &= -H(\nu) \end{aligned}$$

This establishes the fundamental relationship:

$$R(\pi) = \mathrm{KL}(\pi \mid \mu \otimes \nu) - H(\nu)$$

Finally, when applying the infimum over all valid couplings $\pi \in \Pi(\mu, \nu)$ for the Entropic Optimal Transport problem, the target distribution ν is strictly fixed. Consequently, its entropy $H(\nu)$ is a constant independent of the optimization variable π. Minimizing the Optimal Transport cost with $R(\pi)$ is therefore mathematically equivalent to minimizing it with $\mathrm{KL}(\pi \mid \mu \otimes \nu)$, demonstrating that the conditional entropy regularizer inherently behaves as an f-divergence.

$$\mathrm{EOT}_{c,\epsilon}(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \left\{ \int_{\mathcal{X}\times\mathcal{Y}} c(x, y) \mathrm{d}\pi(x, y) + \epsilon R(\pi) \right\}$$

This EOT problem can be understood as a weak OT problem. Let $c : \mathcal{X} \times \mathcal{P}(\mathcal{Y})$ be a weak transport cost, which penalizes the displacement of the a point $x \in \mathcal{X}$ into a distribution $\pi(\cdot \mid x) \in \mathcal{P}(\mathcal{Y})$. Then the weak OT problem is given by:

$$\mathrm{WOT}_c(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathcal{X}} c(x, \pi(\cdot \mid x) \mathrm{d}\mu(x)$$

Let the weak cost be defined as:

$$c_\epsilon(x, \pi(\cdot \mid x)) = \int_{\mathcal{Y}} c(x, y) \mathrm{d}\pi(y \mid x) - \epsilon H(\pi(\cdot \mid x))$$

Then if we substitute this into the WOT problem we derive our original EOT objective:

$$\mathrm{EOT}_{c,\epsilon}(\mu, \nu) = \mathrm{WOT}_{c_\epsilon}(\mu, \nu)$$

This cost is strictly conver and lower semi-continuous, so the solution of the WOT problem exists and it's unique.

Similar to the classical OT, the wek OT problem permits a dual representation. Let $f \in \mathcal{C}(\mathcal{Y})$. We define the weak c-transform $f^c : \mathcal{X} \to \mathbb{R}$ as:

$$f^c(x) = \inf_{q \in \mathcal{P}(\mathcal{Y})} \left\{ c(x, q) - \int_{\mathcal{Y}} f(y) \mathrm{d}q(y) \right\}$$

In our particular case, for the weak cost c_ϵ we have:

$$f^{c_\epsilon}(x) = \inf_{q \in \mathcal{P}(\mathcal{Y})} \left\{ \int_{\mathcal{Y}} c(x, y) \mathrm{d}q(y) - \epsilon H(q) - \int_{\mathcal{Y}} f(y) \mathrm{d}q(y) \right\}$$

We specifically name the objective function as $G_{x,f}(q)$ thus we write the f-transform as:

$$f^{c_\epsilon}(x) = \inf_{q \in \mathcal{P}(\mathcal{Y})} G_{x,f}(q)$$

We assume that $\mathcal{X}$, $\mathcal{Y}$ are compact. So the strong duality holds and we can write the dual objective as (which is equivalent with our original EOT objective):

$$\mathrm{EOT}_{c,\epsilon}(\mu, \nu) = \sup_{f \in \mathcal{C}(\mathcal{Y})} \left\{ \int_{\mathcal{X}} \min_{q_x \in \mathcal{P}(\mathcal{Y})} G_{x,f}(q_x) \mathrm{d}\mu(x) + \int_{\mathcal{Y}} f(y) \mathrm{d}\nu(y) \right\}$$

where the inf because min because we work on compact sets over continuous functions.

Lastly, let's denote the objective functional as $F^w_{c_\epsilon}$ (which is in the class $\mathcal{C}(\mathcal{Y})$), thus we can write the EOT objective in the compact form:

$$\mathrm{EOT}_{c,\epsilon}(\mu, \nu) = \sup_{f \in \mathcal{C}(\mathcal{Y})} F^w_{c_\epsilon}(f)$$

The energy-based models (EBM) parameterize the measure $q \in \mathcal{P}(\mathcal{Y})$ as a Gibbs-Boltzmann distribution density:

$$\frac{\mathrm{d}q(y)}{\mathrm{d}y} = \frac{1}{Z}\exp(-E(y))$$

where $E : \mathcal{Y} \to \mathbb{R}$ is the energy function and $Z = \int_{\mathcal{Y}} \exp(-E(y))\mathrm{d}y$ is the partition function.

Let $q_\theta(y)$ be a parametric family of distributions approximated using a NN which imitates the energy function, meaning:

$$q_\theta(y) = \frac{1}{Z_\theta}\exp(-E_\theta(y))$$

Using q, as the density of the the probability measure $q \in \mathcal{P}(\mathcal{Y})$ w.r.t the Lebesgue measure on $\mathcal{Y}$, the EBMs framework seeks to minimize the KL divergence between the energy representation of q and our neural network:

$$\begin{aligned}
\theta^* &= \arg\min_\theta \mathrm{KL}(q \mid q_\theta) \\
&= \arg\min_\theta \int_{\mathcal{Y}} q(y) \log \frac{q(y)}{q_\theta(y)}\mathrm{d}y \\
&= \arg\min_\theta \left\{ \int_{\mathcal{Y}} q(y)\log q(y)\mathrm{d}y - \int_{\mathcal{Y}} \log q_\theta(y) q(y)\mathrm{d}y \right\} \\
&= \arg\min_\theta \left\{ - \int_{\mathcal{Y}} \log q_\theta(y) q(y)\mathrm{d}y \right\}
\end{aligned}$$

By substituting the energy parameterization of q_θ in the objective we derive:

$$\theta^* = \arg\min_\theta \int_{\mathcal{Y}} (\log Z_\theta + E_\theta(y)) q(y)\mathrm{d}y$$

To find the minimizer we take the derivative of the loss function with respect to the model parameters:

$$\begin{aligned}
\frac{\partial}{\partial\theta}\mathrm{KL}(q \mid q_\theta) &= \int_{\mathcal{Y}} \frac{\partial}{\partial\theta} E_\theta(y)\mathrm{d}q(y) - \int_{\mathcal{Y}} \left[\frac{\partial}{\partial\theta}E_\theta(y)\right]\mathrm{d}q_\theta(y) \\
&= \mathbb{E}_{y\sim q(\cdot)}\left[\frac{\partial}{\partial\theta}E_\theta(y)\right] - \mathbb{E}_{y'\sim q_\theta(\cdot)}\left[\frac{\partial}{\partial\theta}E_\theta(y')\right]
\end{aligned}$$

For the first expectation we sample directly from the terminal distribution $\nu \in \mathcal{P}(\mathcal{Y})$. For the second expectation we use Langevin Dynamics. For an EBM, the score conveniently drops the intractable partition function:

$$\nabla_y \log q_\theta(y) = \nabla_y \left(-E_\theta(y) - \log Z_\theta\right) = -\nabla_y E_\theta(y)$$

To draw samples y' from the model $q_\theta(\cdot)$, we initialize a starting point (often random noise) and iteratively apply the Unadjusted Langevin Algorithm (ULA) update rule:

$$y_{t+1} = y_t - \frac{\eta}{2}\nabla_y E_\theta(y_t) + \sqrt{\eta} z_t, \quad z_t \sim N(0, \mathbf{I})$$

Now, we are going to use the EBM framework in order to solve the EOT objective we have defined about. We will use the three theorems the authors of Energy-Guided Neural Entropic Optimal Transport have proved.

Theorem 1. Let $f \in \mathcal{C}(\mathcal{Y})$, and $x \in \mathcal{X}$. Then the iner weak dual objective $\min_{q\in\mathcal{P}(\mathcal{Y})} G_{x,f}(q)$ permits the unique minimizer q_x^f:

$$\frac{\mathrm{d}q_x^f(y)}{\mathrm{d}y} = \frac{1}{Z(f,x)}\exp\left(\frac{f(y) - c(x,y)}{\epsilon}\right)$$

And for that minimizer the weak c-transform takes the form:

$$\begin{aligned} f^{c_\epsilon}(x) &= \inf_{q\in\mathcal{P}(\mathcal{Y})} G_{x,f}(q) = G_{x,f}(q_x^f) = -\epsilon \log Z(f,x) \\ &= -\epsilon \log \int_{\mathcal{Y}} \exp\left(\frac{f(y) - c(x,y)}{\epsilon}\mathrm{d}y\right) \end{aligned}$$

Therefore, the EOT objective takes the form:

$$\mathrm{EOT}_{c,\epsilon}(\mu,\nu) = \sup_{f\in\mathcal{C}(\mathcal{Y})} \left\{ -\epsilon \int_{\mathcal{X}} \log Z(f,x)\mathrm{d}\mu(x) + \int_{\mathcal{Y}} f(y)\mathrm{d}\nu(y) \right\}$$

where remember that the objective function is defined as $F_{c_\epsilon}^w(f)$.

For a given $f \in \mathcal{C}(\mathcal{Y})$ consider the coupling $\pi^f \in \Pi(\mu,\nu)$ defined as:

$$\pi^f(x,y) = q_x^f(y)\mu(x)$$

Theorem 2. Define the $F_{c_\epsilon}^{w,*} = \mathrm{EOT}_{c,\epsilon}(\mu,\nu)$, and let π^ϵ be the optimal coupling. Then for every $f \in \mathcal{C}(\mathcal{Y})$ it holds that:

$$F_{c_\epsilon}^{w,*} - F_{c_\epsilon}^w(f) = \epsilon \mathrm{KL}(\pi^\epsilon \mid \pi^f)$$

So we parameterize the functions $f \in \mathcal{C}(\mathcal{Y})$ as neural networks and seek to minimize the loss function:

$$L(\theta) = -\epsilon \int_{\mathcal{X}} \log Z(f_\theta, x)\mathrm{d}\mu(x) + \int_{\mathcal{Y}} f_\theta(y)\mathrm{d}\nu(y)$$

Theorem 3. It holds that:

$$\frac{\partial}{\partial\theta}L(\theta) = -\int_{\mathcal{X}\times\mathcal{Y}}\left(\frac{\partial}{\partial\theta}f_\theta(y)\right)\mathrm{d}(\mu\otimes q_x^f)(x,y) + \int_{\mathcal{Y}}\frac{\partial}{\partial\theta}f_\theta(y)\mathrm{d}\nu(y)$$

Algorithm 5: Entropic Optimal Transport via Energy-Based Modelling

Input: source and target distribution μ, ν accessible by samples, entropy regularization coefficient $\epsilon > 0$, cost function $c : \mathcal{X}\times\mathcal{Y} \to \mathbb{R}^+$, number of Langevin steps $k > 0$, Langevin discretization step size $\eta > 0$, basic noise std $\sigma_0 > 0$, potential network $f_\theta : \mathcal{Y} \to \mathbb{R}$, batch size $N > 0$

while *Not Converged* **do**

Sample batches $\{x_n\}_{n=1}^N = X \sim \mu$, $\{y_n\}_{n=1}^N = Y \sim \nu$.

Sample basic noise $Y^{(0)} \sim N(0, \sigma_0)$ of size N.

for $k = 1$ **to** K **do**

Sample $z^{(k)} = \{z_n^{(k)}\}_{n=1}^N$, where $z_n^{(k)} \sim N(0,1)$.

Obtain $Y^{(k)} = \{y_n^{(k)}\}_{n=1}^N$ with Langevin Step:

$$y_n^{(k)} \leftarrow y_n^{(k-1)} + \frac{\eta}{2\epsilon}\text{stop_grad}\left(\frac{\partial}{\partial y}[f_\theta(y) - c(x_n,y)]_{|y=y_n^{(k-1)}}\right) + \sqrt{\eta}z_n^{(k)}$$

$$\hat{L} = -\frac{1}{N}\sum_{y_n^{(k)}\in Y^{(k)}} f_\theta(y_n^{(k)}) + \frac{1}{N}\sum_{y_n\in Y} f_\theta(y_n)$$

Perform gradient step over θ using $\partial_\theta\hat{L}$.

Output: Trained potential network f_θ

Apply EBM to the Schrodinger Bridge Problem

Notation and definitions

Let Ω be the path space and let $\mathcal{M}_+(\Omega)$ be the space of all positive finite measures on Ω. Let $R \in \mathcal{M}_+(\Omega)$ be the reference measure. Define its endpoint marginal $R_{01} \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y})$, such that for any measurable subsets $A \subseteq \mathcal{X}$ and $B \subseteq \mathcal{Y}$, $R_{01}(A\times B) = R((X_0, X_1) \in A\times B)$.

Because the set of couplings $\Pi(\mu,\nu)$ is weakly compact and the Kullback-Leibler divergence is lower semi-continuous, the static Schrödinger Bridge Problem is rigorously

defined as:

$$\pi^* = \arg \min_{\pi \in \Pi(\mu,\nu)} \mathrm{KL}(\pi \mid R_{01})$$

Let $\mathcal{C}_B(\mathcal{X})$ be the space of all continuous and bounded functions on $\mathcal{X}$ (similarly define $\mathcal{C}_B(\mathcal{Y})$). While the theoretical formulation over continuous functions is a supremum, assuming the potentials (ϕ^*, ψ^*) are attained in a suitably relaxed space, the Dual Schrödinger Bridge problem is the following maximization problem:

$$(\phi^*, \psi^*) = \arg\max_{(\phi,\psi)} \left\{ \int_{\mathcal{X}} \phi(x) \mathrm{d}\mu(x) + \int_{\mathcal{Y}} \psi(y) \mathrm{d}\nu(y) - \log \int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi(x) + \psi(y)) \mathrm{d}R_{01}(x,y) \right\}$$

Conditional optimal plan

Let π^* be the solution to the static SP and (ϕ^*, ψ^*) the solutions of the dual SP. The it can be shown that they satisfy:

$$\mathrm{d}\pi^*(x,y) = \frac{\exp(\phi^*(x) + \psi^*(y))}{\int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi^*(x) + \psi^*(y)) \mathrm{d}R_{01}(x,y)} \, \mathrm{d}R_{01}(x,y)$$

By the disintegration theorem, this joint measure can be factored into a marginal measure R_0 on $\mathcal{X}$ and a Markov transition kernel (the conditional measure) $R_{01}(\cdot \mid x)$ on $\mathcal{Y}$. In differential notation, this is commonly written as:

$$\mathrm{d}R_{01}(x,y) = R_{01}(\mathrm{d}y \mid x) \mathrm{d}R_0(x)$$

So we substitute back to the optimal coupling expression:

$$\mathrm{d}\pi^*(x,y) = \frac{\exp(\phi^*(x) + \psi^*(y))}{\int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi^*(x) + \psi^*(y)) R_{01}(\mathrm{d}y \mid x) \mathrm{d}R_0(x)} \, R_{01}(\mathrm{d}y \mid x) \mathrm{d}R_0(x)$$

Also, consider the disintegration of the optimal coupling as a marginal measure $\pi^*(x) = \mu(x)$ on $\mathcal{X}$ and a conditional measure $\pi^*(\cdot \mid x)$ on $\mathcal{Y}$:

$$\mathrm{d}\pi^*(x,y) = \pi^*(\mathrm{d}y \mid x) \mathrm{d}\mu(x)$$

Therefore if we substitute the analytic expression of the differential of the optimal coupling and integrate over the entire space $\mathcal{Y}$ we derive:

$$\mathrm{d}\mu(x) = \frac{\exp(\phi^*(x)) \int_{\mathcal{Y}} \exp(\psi^*(y)) R_{01}(\mathrm{d}y \mid x)}{\int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi^*(x) + \psi^*(y)) R_{01}(\mathrm{d}y \mid x) \mathrm{d}R_0(x)} \mathrm{d}R_0(x)$$

This allow us to write:

$$\frac{1}{\int_{\mathcal{Y}} \exp(\psi^*(y)) R_{01}(\mathrm{d}y \mid x)} \, \mathrm{d}\mu(x) = \frac{\exp(\phi^*(x))}{\int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi^*(x) + \psi^*(y)) R_{01}(\mathrm{d}y \mid x) \mathrm{d}R_0(x)} \mathrm{d}R_0(x)$$

Which help us simplify the expression of the differential of the optimal coupling:

$$\mathrm{d}\pi^*(x, y) = \frac{\exp(\psi^*(y)) R_{01}(\mathrm{d}y \mid x)}{\int_{\mathcal{Y}} \exp(\psi^*(y)) R_{01}(\mathrm{d}y \mid x)} \mathrm{d}\mu(x)$$

So, lastly if we also substitute the disintegration of $\pi^*(x, y)$ we derive the closed form of the conditional optimal plan:

$$\pi^*(\mathrm{d}y \mid x) = \frac{\exp(\psi^*(y))}{\int_{\mathcal{Y}} \exp(\psi^*(y)) R_{01}(\mathrm{d}y \mid x)} \, R_{01}(\mathrm{d}y \mid x)$$

Semi-dual objective

Now, we will construct the objective that yields this conditional optimal plan. Recall the full objective function of the dual problem:

$$J(\phi, \psi) = \int_{\mathcal{X}} \phi(x) \mathrm{d}\mu(x) + \int_{\mathcal{Y}} \psi(y) \mathrm{d}\nu(y) - \log \int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi(x) + \psi(y)) \mathrm{d}R_{01}(x, y)$$

To transition from the full dual to the semi-dual objective rigorously, we rely on the property of partial maximization. Instead of optimizing both potentials simultaneously, we nest the supremum:

$$\sup_{\phi,\psi} J(\phi, \psi) = \sup_{\psi} \left(\sup_{\phi} J(\phi, \psi) \right)$$

By fixing the target potential ψ, we can first solve the inner maximization to find the conditionally optimal source potential, denoted ϕ^ψ (the entropic c-transform). We utilize the calculus of variations to rigorously derive the stationary condition for this functional. We perturb the optimal potential by an arbitrary continuous and bounded test function $h \in \mathcal{C}_B(\mathcal{X})$ scaled by a real parameter ϵ:

$$\phi_\epsilon(x) = \phi^\psi(x) + \epsilon h(x)$$

The first-order necessary condition for maximization requires the Gâteaux derivative of J with respect to ϵ, evaluated at $\epsilon = 0$, to be exactly zero for all possible test functions

$h(x)$:

$$\frac{d}{d\epsilon}J(\phi_\epsilon,\psi)\bigg|_{\epsilon=0} = \frac{d}{d\epsilon}\left[\int_{\mathcal{X}}(\phi^\psi(x)+\epsilon h(x))\mathrm{d}\mu(x) - \log\int_{\mathcal{X}\times\mathcal{Y}}\exp(\phi^\psi(x)+\epsilon h(x)+\psi(y))\mathrm{d}R_{01}(x,y)\right]\bigg|_{\epsilon=0}$$

$$= \int_{\mathcal{X}} h(x)\mathrm{d}\mu(x) - \left.\frac{\int_{\mathcal{X}\times\mathcal{Y}} h(x)\exp(\phi^\psi(x)+\epsilon h(x)+\psi(y))\mathrm{d}R_{01}(x,y)}{\int_{\mathcal{X}\times\mathcal{Y}}\exp(\phi^\psi(x')+\epsilon h(x')+\psi(y))\mathrm{d}R_{01}(x',y)}\right|_{\epsilon=0}$$

$$= \int_{\mathcal{X}} h(x)\mathrm{d}\mu(x) - \frac{\int_{\mathcal{X}\times\mathcal{Y}} h(x)\exp(\phi^\psi(x)+\psi(y))\mathrm{d}R_{01}(x,y)}{\int_{\mathcal{X}\times\mathcal{Y}}\exp(\phi^\psi(x')+\psi(y))\mathrm{d}R_{01}(x',y)} = 0$$

To isolate $h(x)$, we apply the disintegration theorem to the reference measure in the numerator, $\mathrm{d}R_{01}(x,y) = R_{01}(\mathrm{d}y \mid x)\mathrm{d}R_0(x)$, and factor out the terms depending only on x:

$$\int_{\mathcal{X}} h(x)\mathrm{d}\mu(x) = \int_{\mathcal{X}} h(x)\left[\frac{\exp(\phi^\psi(x))\int_{\mathcal{Y}}\exp(\psi(y))R_{01}(\mathrm{d}y \mid x)}{\int_{\mathcal{X}\times\mathcal{Y}}\exp(\phi^\psi(x')+\psi(y))\mathrm{d}R_{01}(x',y)}\right]\mathrm{d}R_0(x)$$

By the fundamental lemma of the calculus of variations, because this equality holds for *any* arbitrary test function $h(x)$, the underlying measures inside the integrals must be strictly equal almost everywhere:

$$\mathrm{d}\mu(x) = \frac{\exp(\phi^\psi(x))\int_{\mathcal{Y}}\exp(\psi(y))R_{01}(\mathrm{d}y \mid x)}{\int_{\mathcal{X}\times\mathcal{Y}}\exp(\phi^\psi(x')+\psi(y))\mathrm{d}R_{01}(x',y)}\,\mathrm{d}R_0(x)$$

Finally, by dividing both sides by the conditional integral over $\mathcal{Y}$ (which is strictly positive), we recover the exact stationary condition:

$$\frac{1}{\int_{\mathcal{Y}}\exp(\psi(y))R_{01}(\mathrm{d}y \mid x)}\,\mathrm{d}\mu(x) = \frac{\exp(\phi^\psi(x))}{\int_{\mathcal{X}\times\mathcal{Y}}\exp(\phi^\psi(x')+\psi(y))\mathrm{d}R_{01}(x',y)}\mathrm{d}R_0(x)$$

To explicitly solve for $\phi^\psi(x)$, we isolate the exponential term by rearranging the stationary condition. Let $Z_{\phi,\psi} = \int_{\mathcal{X}\times\mathcal{Y}}\exp(\phi^\psi(x')+\psi(y))\mathrm{d}R_{01}(x',y)$ denote the joint partition function, which is a global scalar constant for any fixed pair of potentials. The condition becomes:

$$\exp(\phi^\psi(x)) = \frac{\mathrm{d}\mu(x)}{\mathrm{d}R_0(x)}\,\frac{Z_{\phi,\psi}}{\int_{\mathcal{Y}}\exp(\psi(y))R_{01}(\mathrm{d}y \mid x)}$$

Taking the natural logarithm of both sides yields the closed-form expression for the optimal source potential:

$$\phi^\psi(x) = \log\left(\frac{\mathrm{d}\mu(x)}{\mathrm{d}R_0(x)}\right) - \log\left(\int_{\mathcal{Y}}\exp(\psi(y))R_{01}(\mathrm{d}y \mid x)\right) + \log Z_{\phi,\psi}$$

We now substitute this conditionally optimal $\phi^{\psi}(x)$ back into the full dual objective $J(\phi^{\psi}, \psi)$. We can evaluate the integral over $\mathcal{X}$ term by term:

$$\int_{\mathcal{X}} \phi^{\psi}(x)\mathrm{d}\mu(x) = \int_{\mathcal{X}} \log\left(\frac{\mathrm{d}\mu(x)}{\mathrm{d}R_0(x)}\right)\mathrm{d}\mu(x) - \int_{\mathcal{X}} \log\left(\int_{\mathcal{Y}} \exp(\psi(y))R_{01}(\mathrm{d}y \mid x)\right)\mathrm{d}\mu(x) + \int_{\mathcal{X}} \log Z_{\phi,\psi}\mathrm{d}\mu(x)$$

The first term is exactly the definition of the Kullback-Leibler divergence between the source data distribution and the reference source marginal, $\mathrm{KL}(\mu \mid R_0)$. For the third term, because $\log Z_{\phi,\psi}$ is a constant with respect to x and μ is a valid probability measure ($\int_{\mathcal{X}} \mathrm{d}\mu(x) = 1$), it simplifies to strictly $\log Z_{\phi,\psi}$.

Reconstructing the full objective $J(\phi^{\psi}, \psi)$, we observe a perfect cancellation of the partition function. The $+\log Z_{\phi,\psi}$ generated by integrating the source potential exactly cancels the $-\log Z_{\phi,\psi}$ native to the third term of the dual objective:

$$\begin{aligned} J(\phi^{\psi}, \psi) &= \int_{\mathcal{X}} \phi^{\psi}(x)\mathrm{d}\mu(x) + \int_{\mathcal{Y}} \psi(y)\mathrm{d}\nu(y) - \log Z_{\phi,\psi} \\ &= \mathrm{KL}(\mu \mid R_0) - \int_{\mathcal{X}} \log\left(\int_{\mathcal{Y}} \exp(\psi(y))R_{01}(\mathrm{d}y \mid x)\right)\mathrm{d}\mu(x) + \log Z_{\phi,\psi} \\ &\quad + \int_{\mathcal{Y}} \psi(y)\mathrm{d}\nu(y) - \log Z_{\phi,\psi} \end{aligned}$$

This leaves the completely reduced semi-dual objective, heavily parameterized by ψ alone:

$$J(\psi) = \int_{\mathcal{Y}} \psi(y)\mathrm{d}\nu(y) - \int_{\mathcal{X}} \log\left(\int_{\mathcal{Y}} \exp(\psi(y))R_{01}(\mathrm{d}y \mid x)\right)\mathrm{d}\mu(x) + \mathrm{KL}(\mu \mid R_0)$$

By discarding the constant divergence term $\mathrm{KL}(\mu \mid R_0)$, the problem of finding the optimal target potential ψ^* formally reduces to the following unconstrained maximization problem:

$$\psi^* = \underset{\psi \in \mathcal{C}_B(\mathcal{Y})}{\arg\max} \left\{ \int_{\mathcal{Y}} \psi(y)\mathrm{d}\nu(y) - \int_{\mathcal{X}} \log\left(\int_{\mathcal{Y}} \exp(\psi(y))R_{01}(\mathrm{d}y \mid x)\right)\mathrm{d}\mu(x) \right\}$$

Crafting the learning algorithm

To construct a tractable learning algorithm, we parameterize the continuous target potential ψ using a neural network with weights θ, denoted as ψ_θ. The optimization

problem thus becomes finding the parameters θ that maximize the semi-dual objective $J(\theta)$:

$$J(\theta) = \int_{\mathcal{Y}} \psi_\theta(y) \mathrm{d}\nu(y) - \int_{\mathcal{X}} \log \left(\int_{\mathcal{Y}} \exp(\psi_\theta(y)) R_{01}(\mathrm{d}y \mid x) \right) \mathrm{d}\mu(x)$$

We maximize this objective via gradient ascent. Taking the derivative of $J(\theta)$ with respect to the network parameters θ, we apply the gradient operator linearly to both terms:

$$\nabla_\theta J(\theta) = \int_{\mathcal{Y}} \nabla_\theta \psi_\theta(y) \mathrm{d}\nu(y) - \int_{\mathcal{X}} \nabla_\theta \log \left(\int_{\mathcal{Y}} \exp(\psi_\theta(y)) R_{01}(\mathrm{d}y \mid x) \right) \mathrm{d}\mu(x)$$

For the second term, we apply the chain rule to the natural logarithm ($\nabla \log f = \frac{\nabla f}{f}$). We can also safely pass the derivative operator inside the integral over $\mathcal{Y}$ under standard regularity conditions:

$$\nabla_\theta J(\theta) = \int_{\mathcal{Y}} \nabla_\theta \psi_\theta(y) \mathrm{d}\nu(y) - \int_{\mathcal{X}} \left[\frac{\int_{\mathcal{Y}} \nabla_\theta \psi_\theta(y) \exp(\psi_\theta(y)) R_{01}(\mathrm{d}y \mid x)}{\int_{\mathcal{Y}} \exp(\psi_\theta(y')) R_{01}(\mathrm{d}y' \mid x)} \right] \mathrm{d}\mu(x)$$

Notice that the fractional term inside the bracket perfectly reconstructs the definition of the conditional optimal plan $\pi^*_\theta(\mathrm{d}y \mid x)$ we derived earlier. By substituting the measure $\pi^*_\theta(\mathrm{d}y \mid x)$ back into the equation, we obtain:

$$\nabla_\theta J(\theta) = \int_{\mathcal{Y}} \nabla_\theta \psi_\theta(y) \mathrm{d}\nu(y) - \int_{\mathcal{X}} \left[\int_{\mathcal{Y}} \nabla_\theta \psi_\theta(y) \pi^*_\theta(\mathrm{d}y \mid x) \right] \mathrm{d}\mu(x)$$

Finally, rewriting these integrals as expectations over their respective probability measures yields the exact training gradient:

$$\nabla_\theta J(\theta) = \mathbb{E}_{y \sim \nu} \left[\nabla_\theta \psi_\theta(y) \right] - \mathbb{E}_{x \sim \mu} \left[\mathbb{E}_{y' \sim \pi^*_\theta(\cdot \mid x)} \left[\nabla_\theta \psi_\theta(y') \right] \right]$$

Evaluating this gradient poses a practical computational challenge. While the first expectation over ν can be easily approximated by drawing batches of real target data, the inner expectation of the second term requires sampling from the conditional optimal plan:

$$\pi^*_\theta(y \mid x) = \frac{\exp(\psi_\theta(y)) R_{01}(y \mid x)}{\int_{\mathcal{Y}} \exp(\psi_\theta(y')) R_{01}(y' \mid x) \mathrm{d}y'}$$

Because the target potential ψ_θ is parameterized by a highly non-linear neural network, the normalizing integral in the denominator (the partition function) is intractable. Consequently, direct sampling from $\pi^*_\theta(\cdot \mid x)$ is impossible. To overcome this, we turn

to Langevin dynamics, a Markov Chain Monte Carlo (MCMC) method that allows us to sample from a distribution utilizing only its unnormalized log-density.

To implement this in practice, we apply the Euler-Maruyama discretization, yielding the Unadjusted Langevin Algorithm (ULA). Given our specific unnormalized target density $\pi_\theta^*(y \mid x) \propto \exp(\psi_\theta(y) + \log R_{01}(y \mid x))$, the discrete iterative update rule for a particle at step k with a step size $\eta > 0$ becomes:

$$y^{(k)} = y^{(k-1)} + \frac{\eta}{2}\nabla_y \left[\psi_\theta(y^{(k-1)}) + \log R_{01}(y^{(k-1)} \mid x)\right] + \sqrt{\eta} z^{(k)}$$

where $z^{(k)} \sim \mathcal{N}(0, I)$ is standard Gaussian noise. By simulating this chain for K steps, the empirical distribution of the particles asymptotically approximates the true conditional distribution $\pi_\theta^*(\cdot \mid x)$, enabling the approximation of the intractable negative phase in our training gradient.

Algorithm 6: Neural Static Schrödinger Bridge via Energy-Based Modeling

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Input: source and target distributions μ, ν accessible by samples, reference conditional measure $R_{01}(y \mid x)$, number of Langevin steps $K > 0$, Langevin step size $\eta > 0$, target potential network $\psi_\theta : \mathcal{Y} \to \mathbb{R}$, batch size $N > 0$, learning rate $\gamma > 0$

while *Not Converged* **do**

Sample real data batches $\{x_n\}_{n=1}^N = X \sim \mu$, and $\{y_n\}_{n=1}^N = Y \sim \nu$.

Initialize fake samples from the reference conditional: $Y^{(0)} = \{y_n^{(0)}\}_{n=1}^N$, where $y_n^{(0)} \sim R_{01}(\cdot \mid x_n)$.

for $k = 1$ **to** K **do**

Sample standard noise $z^{(k)} = \{z_n^{(k)}\}_{n=1}^N$, where $z_n^{(k)} \sim \mathcal{N}(0, I)$.

Obtain $Y^{(k)} = \{y_n^{(k)}\}_{n=1}^N$ using the Unadjusted Langevin Algorithm (ULA):

$$y_n^{(k)} \leftarrow y_n^{(k-1)} + \frac{\eta}{2}\text{stop_grad}\left(\frac{\partial}{\partial y}[\psi_\theta(y) + \log R_{01}(y \mid x_n)]\Big|_{y=y_n^{(k-1)}}\right) + \sqrt{\eta} z_n^{(k)}$$

Compute the empirical semi-dual loss (to be minimized):

$$\hat{L} = \frac{1}{N}\sum_{y_n^{(K)} \in Y^{(K)}} \psi_\theta(y_n^{(K)}) - \frac{1}{N}\sum_{y_n \in Y} \psi_\theta(y_n)$$

Perform gradient descent step over the network parameters θ:

$$\theta \leftarrow \theta - \gamma \nabla_\theta \hat{L}(\theta)$$

Output: Trained target potential network ψ_θ

Crafting the sampling algorithm

Once the target potential ψ_θ is fully trained, mapping a new source sample $x \sim \mu$ to the target distribution requires sampling from the learned conditional plan $\pi^*_\theta(\cdot \mid x)$. This is achieved by running the Langevin diffusion process without computing parameter gradients. To ensure high-fidelity generation, the inference chain is typically simulated for a large number of steps K_{infer} to guarantee convergence to the stationary distribution.

Algorithm 7: Generation via Learned Schrödinger Bridge

Input: source sample $x \sim \mu$, trained target potential network ψ_θ, reference noise scale $\epsilon > 0$, number of inference Langevin steps K_{infer}, Langevin step size $\eta > 0$

Initialize the particle directly from the reference conditional prior (e.g., standard Brownian transition):

$$y^{(0)} \sim \mathcal{N}(x, \epsilon I)$$

for $k = 1$ **to** K_{infer} **do**

Sample standard Gaussian noise $z^{(k)} \sim \mathcal{N}(0, I)$.

Evaluate the learned vector field (drift) at the current state:

$$v(y^{(k-1)}, x) = \nabla_y[\psi_\theta(y) + \log R_{01}(y \mid x_n)]\Big|_{y=y_n^{(k-1)}}$$

Update the particle state using the Unadjusted Langevin Algorithm:

$$y^{(k)} \leftarrow y^{(k-1)} + \frac{\eta}{2} v(y^{(k-1)}, x) + \sqrt{\eta} z^{(k)}$$

Output: Generated target sample $y^{(K_{\text{infer}})}$ distributed according to $\pi^*_\theta(\cdot \mid x)$

A fundamental theoretical property of the dynamic Schrödinger problem is that the optimal dynamic path measure, P^*, minimizes the relative entropy to R by altering only the marginal distribution of the endpoints, leaving the conditional path dynamics identical to the prior. Therefore, the optimal dynamic measure disintegrates using the exact same Brownian Bridge measure R^{xy}, but integrated over the optimal static plan π^*:

$$P^*(\mathrm{d}\omega) = \int_{\mathcal{X}\times\mathcal{Y}} R^{xy}(\mathrm{d}\omega)\mathrm{d}\pi^*(x, y)$$

where $R^{xy}(\cdot) = R(\cdot \mid X_0 = x, X_1 = y)$ is the strictly Markovian Brownian Bridge connecting x to y in unit time if R is a Brownian Motion measure.

Consequently, to sample a continuous trajectory $\omega \sim P^*$ spanning $t \in [0, 1]$, one does

not need to learn a time-dependent vector field. The generation process strictly decomposes into two sequential steps:

1. Endpoint Sampling: Draw a source sample $x \sim \mu$, and use the trained Energy-Based Model (via the Unadjusted Langevin Algorithm) to draw the corresponding target endpoint $y \sim \pi_\theta^*(\cdot \mid x)$.

2. Trajectory Simulation: Given the fixed endpoints (x, y), sample the intermediate path from the standard Brownian Bridge measure R^{xy}.

For example, in the continuous-time Brownian Bridge from x to y with noise scale ϵ, it is governed by the following SDE for $t \in [0, 1)$:

$$\mathrm{d}X_t = \frac{y - X_t}{1 - t}\mathrm{d}t + \sqrt{\epsilon}\mathrm{d}B_t, \quad X_0 = x$$

By simulating this SDE computationally (e.g., via Euler-Maruyama discretization), we construct the exact physical trajectories of the Schrödinger Bridge without requiring any time-parameterized neural networks.

Algorithmic implementations

1.

In standard continuous formulations of the Schrödinger Bridge, the reference process is defined as a scaled Brownian motion governing the prior dynamics. If particles evolve according to the Stochastic Differential Equation $\mathrm{d}X_t = \sqrt{\epsilon}\mathrm{d}B_t$, where $\epsilon > 0$ dictates the noise scale, the transition probability from an initial state x at $t = 0$ to a state y at $t = 1$ is strictly Gaussian.

The conditional density of this reference measure over $\mathbb{R}^d$ is given by:

$$R_{01}(y \mid x) = \frac{1}{(2\pi\epsilon)^{d/2}} \exp\left(-\frac{\|x - y\|^2}{2\epsilon}\right)$$

Taking the natural logarithm of this density isolates the Euclidean distance between the source and target points:

$$\log R_{01}(y \mid x) = -\frac{1}{2\epsilon}\|x - y\|^2 - \frac{d}{2}\log(2\pi\epsilon)$$

Notice that by defining the optimal transport cost function as the squared Euclidean distance, $c(x, y) = \frac{1}{2}\|x - y\|^2$, the log-prior directly corresponds to the entropic penalty

term $-\frac{1}{\epsilon}c(x, y)$. The additive constant drops out during differentiation. Consequently, the gradient of the log-prior with respect to the target variable y is purely linear:

$$\nabla_y \log R_{01}(y \mid x) = -\frac{1}{\epsilon}(y - x) = \frac{x - y}{\epsilon}$$

Substituting this closed-form gradient into the Unadjusted Langevin Algorithm (ULA), the iterative update rule for sampling from the conditional optimal plan simplifies significantly:

$$y^{(k)} = y^{(k-1)} + \frac{\eta}{2}\text{stop_grad}\left(\nabla_y \psi_\theta(y^{(k-1)}) + \frac{x - y^{(k-1)}}{\epsilon}\right) + \sqrt{\eta} z^{(k)}$$

2.

In the ULA update step, the stop_grad operator acts as a crucial computational constraint. Mathematically, the semi-dual objective requires us to evaluate the expected gradient of the potential over samples drawn from the conditional distribution $\pi^*_\theta(\cdot \mid x)$. The Langevin dynamics serve strictly as an independent sampling subroutine to approximate this expectation. If the gradient operator is not detached during this simulation, automatic differentiation frameworks will erroneously backpropagate the outer parameter updates ∇_θ through the inner iterative sampling path, computing unintended higher-order derivatives ($\nabla_\theta \nabla_y \psi_\theta$). Detaching the graph ensures memory efficiency and preserves the mathematical integrity of the Contrastive Divergence gradient step.

3.

While the presented algorithms are theoretically rigorous for any fixed entropy penalty $\epsilon > 0$, optimizing the Energy-Based Model in practice exposes an intrinsic mathematical trade-off between transport accuracy and MCMC mixing. As $\epsilon \to 0$, the entropic transport plan sharply approaches the true deterministic Monge map. However, as a direct consequence, the conditional reference measure $R_{01}(y \mid x) = \mathcal{N}(y; x, \epsilon I)$ collapses into a highly concentrated Dirac spike around the source point x.

Computationally, this concentration causes the Unadjusted Langevin Algorithm (ULA) to suffer from severe poor mixing during the negative phase of the Contrastive Divergence loop. If the target distribution ν is spatially distant from the source distribution μ, the Langevin chains initialized at x will fail to explore the state space and reach the target modes within a finite K steps. This dynamic is commonly known as the cold-start problem, which results in uninformative training gradients and catastrophic mode collapse.

Consequently, while the semi-dual derivation remains mathematically unchanged, stabilizing the training algorithm in practice strictly requires integrating standard EBM engineering techniques:

- Langevin Replay Buffers: Instead of strictly initializing $Y^{(0)}$ from the reference prior at every iteration, the algorithm maintains a persistent memory bank of previously generated samples. Resuming the Markov chain from these buffered states bridges the spatial gap and ensures stable traversal of the target domain.
- Simulated ϵ-Annealing: Training is initialized with a large ϵ, allowing the reference measure to act as a wide prior that facilitates global exploration. The value of ϵ is then deterministically decayed over the course of training, dynamically tightening the transport plan only after the network has learned the macroscopic topological mapping.

Algorithmic Variation Based on SCONE

Detailing Approach

While the semi-dual formulation analytically eliminates the source potential, one can alternatively attempt to solve the full dual problem by parameterizing and learning both potentials ϕ_{θ_1} and ψ_{θ_2} simultaneously. However, evaluating the joint dynamics in this unconstrained space requires an additional pre-training phase: regressing a neural network $s_\omega(x, y)$ to approximate the intractable score of the prior, $\nabla_{x,y} \log R_{01}(x, y)$, using Denoising Score Matching.

To rigorously justify this requirement, recall the objective functional of the full dual representation:

$$J(\phi_{\theta_1}, \psi_{\theta_2}) = \int_{\mathcal{X}} \phi_{\theta_1}(x) \mathrm{d}\mu(x) + \int_{\mathcal{Y}} \psi_{\theta_2}(y) \mathrm{d}\nu(y) - \log \int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi_{\theta_1}(x) + \psi_{\theta_2}(y)) \mathrm{d}R_{01}(x, y)$$

Taking the gradient of this functional with respect to the source network parameters θ_1 yields:

$$\begin{aligned}\nabla_{\theta_1} J(\phi_{\theta_1}, \psi_{\theta_2}) = &\int_{\mathcal{X}} \nabla_{\theta_1} \phi_{\theta_1}(x) \mathrm{d}\mu(x) \\ &- \int_{\mathcal{X}\times\mathcal{Y}} \nabla_{\theta_1} \phi_{\theta_1}(x) \left[\frac{\exp(\phi_{\theta_1}(x) + \psi_{\theta_2}(y))}{\int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi_{\theta_1}(x') + \psi_{\theta_2}(y')) \mathrm{d}R_{01}(x', y')} \right] \mathrm{d}R_{01}(x, y)\end{aligned}$$

The bracketed term perfectly reconstructs the analytical form of the optimal transport plan. By defining the current neural approximation of the joint coupling as $\pi_{\theta_1\theta_2}$:

$$\mathrm{d}\pi_{\theta_1\theta_2}(x, y) = \frac{\exp(\phi_{\theta_1}(x) + \psi_{\theta_2}(y))}{\int_{\mathcal{X}\times\mathcal{Y}} \exp(\phi_{\theta_1}(x') + \psi_{\theta_2}(y')) \mathrm{d}R_{01}(x', y')} \, \mathrm{d}R_{01}(x, y)$$

We substitute this parameterized measure back into the gradient. Because the operator $\nabla_{\theta_1}\phi_{\theta_1}(x)$ depends exclusively on x, we can integrate out y from the joint measure to form the source marginal $\pi^X_{\theta_1\theta_2}(\mathrm{d}x)$:

$$\begin{aligned}\nabla_{\theta_1} J(\phi_{\theta_1}, \psi_{\theta_2}) &= \int_{\mathcal{X}} \nabla_{\theta_1}\phi_{\theta_1}(x)\mathrm{d}\mu(x) - \int_{\mathcal{X}\times\mathcal{Y}} \nabla_{\theta_1}\phi_{\theta_1}(x)\mathrm{d}\pi_{\theta_1\theta_2}(x,y) \\ &= \int_{\mathcal{X}} \nabla_{\theta_1}\phi_{\theta_1}(x)\mathrm{d}\mu(x) - \int_{\mathcal{X}} \nabla_{\theta_1}\phi_{\theta_1}(x)\pi^X_{\theta_1\theta_2}(\mathrm{d}x)\end{aligned}$$

Rewriting this using expectation notation cleanly reveals the Contrastive Divergence structure. By symmetry, the exact same derivation applies to the target potential parameters θ_2:

$$\nabla_{\theta_1} J(\phi_{\theta_1}, \psi_{\theta_2}) = \mathbb{E}_{x\sim\mu(\cdot)}[\nabla_{\theta_1}\phi_{\theta_1}(x)] - \mathbb{E}_{(x,y)\sim\pi_{\theta_1\theta_2}(\cdot,\cdot)}[\nabla_{\theta_1}\phi_{\theta_1}(x)]$$

$$\nabla_{\theta_2} J(\phi_{\theta_1}, \psi_{\theta_2}) = \mathbb{E}_{y\sim\nu(\cdot)}[\nabla_{\theta_2}\psi_{\theta_2}(y)] - \mathbb{E}_{(x,y)\sim\pi_{\theta_1\theta_2}(\cdot,\cdot)}[\nabla_{\theta_2}\psi_{\theta_2}(y)]$$

Evaluating these gradients requires drawing joint samples (x, y) from the optimal coupling $\pi_{\theta_1\theta_2}$. Because the normalizing constant is intractable, we must simulate this joint distribution using Langevin dynamics over the concatenated state space $z = (x, y) \in \mathcal{X} \times \mathcal{Y}$.

The unnormalized log-density of the joint target distribution is given by:

$$\log \pi_{\theta_1\theta_2}(x, y) = \phi_{\theta_1}(x) + \psi_{\theta_2}(y) + \log R_{01}(x, y) + C, \quad C \sim \text{const. w.r.t x,y}$$

To construct the drift vector field for the Langevin diffusion, we take the gradient of this joint log-density with respect to the spatial coordinates x and y. Because ϕ_{θ_1} depends only on x and ψ_{θ_2} depends only on y, the score function elegantly decouples into two partial derivatives:

$$\nabla_{x,y} \log \pi_{\theta_1\theta_2}(x, y) = \begin{bmatrix}\nabla_x \phi_{\theta_1}(x) \\ \nabla_y \psi_{\theta_2}(y)\end{bmatrix} + \nabla_{x,y} \log R_{01}(x, y)$$

Here, the fundamental challenge of the arbitrary reference measure emerges: the term $\nabla_{x,y} \log R_{01}(x, y)$ is intractable. This necessitates the pre-trained score network:

$$s_\omega(x, y) = [s^x_\omega(x, y), s^y_\omega(x, y)]^\top \approx \nabla_{x,y} \log R_{01}(x, y)$$

which is obtained via Denoising Score Matching.

By substituting the neural score network into the joint gradient, we can explicitly define the Unadjusted Langevin Algorithm (ULA) for the full dual. Given a step size $\eta > 0$

and independent standard Gaussian noise $z_x^{(k)}, z_y^{(k)} \sim \mathcal{N}(0, I)$, the joint state updates iteratively as:

$$x^{(k)} = x^{(k-1)} + \frac{\eta}{2}\left[\nabla_x \phi_{\theta_1}(x^{(k-1)}) + s_\omega^x(x^{(k-1)}, y^{(k-1)})\right] + \sqrt{\eta} z_x^{(k)}$$

$$y^{(k)} = y^{(k-1)} + \frac{\eta}{2}\left[\nabla_y \psi_{\theta_2}(y^{(k-1)}) + s_\omega^y(x^{(k-1)}, y^{(k-1)})\right] + \sqrt{\eta} z_y^{(k)}$$

By initializing $(x^{(0)}, y^{(0)}) \sim R_{01}$ and running this coupled diffusion for K steps, the generated particles asymptotically approximate the joint measure $\pi_{\theta_1\theta_2}$, allowing for the evaluation of the full dual Contrastive Divergence gradients.

Algorithm 8: Pre-training the Reference Score via Denoising Score Matching

Input: reference measure R_{01} accessible by samples, noise schedule $\tau_1 > \tau_2 > \cdots > \tau_N > 0$, joint score network $s_\omega = [s_\omega^x, s_\omega^y]^\top$, batch size B, learning rate $\gamma' > 0$

while *Not Converged* **do**

Sample a batch of clean joint pairs $\{(x_b, y_b)\}_{b=1}^B \sim R_{01}$.
Sample noise levels $\{\tau_b\}_{b=1}^B$ uniformly from the schedule $\{\tau_1, \ldots, \tau_N\}$.
Sample standard Gaussian noise $\{\epsilon_b^x\}_{b=1}^B, \{\epsilon_b^y\}_{b=1}^B \sim \mathcal{N}(0, I)$.
For each sample in the batch, generate the noise-perturbed states:

$$\tilde{x}_b = x_b + \sqrt{\tau_b}\epsilon_b^x$$
$$\tilde{y}_b = y_b + \sqrt{\tau_b}\epsilon_b^y$$

Compute the target score for the Gaussian perturbation, $-\frac{\epsilon}{\sqrt{\tau}}$, and evaluate the τ-weighted Mean Squared Error (to equalize gradient variance across noise levels):

$$\mathcal{L}(\omega) = \frac{1}{B}\sum_{b=1}^{B} \tau_b \left(\left\| s_\omega^x(\tilde{x}_b, \tilde{y}_b, \tau_b) + \frac{\epsilon_b^x}{\sqrt{\tau_b}} \right\|_2^2 + \left\| s_\omega^y(\tilde{x}_b, \tilde{y}_b, \tau_b) + \frac{\epsilon_b^y}{\sqrt{\tau_b}} \right\|_2^2 \right)$$

Perform a gradient descent step over the score network parameters ω:

$$\omega \leftarrow \omega - \gamma \nabla_\omega \mathcal{L}(\omega)$$

Output: Trained noise-conditioned score network $s_\omega \approx \nabla_{x,y} \log R_{01}^\tau$

Algorithm 9: Full Dual Schrödinger Bridge

Input: source and target distributions μ, ν, reference measure R_{01} accessible by samples, pre-trained noise-conditioned joint score network $s_\omega(x, y, \tau) \approx \nabla_{x,y} \log R_{01}^\tau(x, y)$, noise schedule $\tau_1 > \tau_2 > \cdots > \tau_N > 0$, Langevin steps K, base step size $\eta > 0$, dual potential networks $\phi_{\theta_1}, \psi_{\theta_2}$, batch size B, learning rate $\gamma > 0$

while *Not Converged* **do**

Sample real data batches $\{x_b\}_{b=1}^B \sim \mu$ and $\{y_b\}_{b=1}^B \sim \nu$.

Initialize fake joint samples from the reference measure: $(X^{(0)}, Y^{(0)}) \sim R_{01}$.

for $i = 1$ **to** N *(Noise Annealing Schedule)* **do**

Set scaled step size $\eta_i = \eta \cdot (\tau_i/\tau_1)$.

for $k = 1$ **to** K *(Langevin Steps)* **do**

Sample standard Gaussian noise $z_x^{(k)}, z_y^{(k)} \sim \mathcal{N}(0, I)$.

$$\Delta_x^{(k)} = \nabla_x \phi_{\theta_1}(X^{(k-1)}) + s_\omega^x(X^{(k-1)}, Y^{(k-1)}, \tau_i)$$

$$\Delta_y^{(k)} = \nabla_y \psi_{\theta_2}(Y^{(k-1)}) + s_\omega^y(X^{(k-1)}, Y^{(k-1)}, \tau_i)$$

$$X^{(k)} \leftarrow X^{(k-1)} + \frac{\eta_i}{2}\text{stop_grad}(\Delta_x^{(k)}) + \sqrt{\eta_i} z_x^{(k)}$$

$$Y^{(k)} \leftarrow Y^{(k-1)} + \frac{\eta_i}{2}\text{stop_grad}(\Delta_y^{(k)}) + \sqrt{\eta_i} z_y^{(k)}$$

$X^{(0)} \leftarrow X^{(K)}$; $Y^{(0)} \leftarrow Y^{(K)}$

Let $(\tilde{X}, \tilde{Y})$ denote the final generated batch of joint particles.

Compute empirical gradients for both potentials (Contrastive Divergence):

$$\nabla_{\theta_1} \hat{J} = \frac{1}{B}\sum_{b=1}^B \nabla_{\theta_1} \phi_{\theta_1}(x_b) - \frac{1}{B}\sum_{b=1}^B \nabla_{\theta_1} \phi_{\theta_1}(\tilde{X}_b)$$

$$\nabla_{\theta_2} \hat{J} = \frac{1}{B}\sum_{b=1}^B \nabla_{\theta_2} \psi_{\theta_2}(y_b) - \frac{1}{B}\sum_{b=1}^B \nabla_{\theta_2} \psi_{\theta_2}(\tilde{Y}_b)$$

Update network parameters to maximize the dual objective:

$$\theta_1 \leftarrow \theta_1 + \gamma \nabla_{\theta_1} \hat{J}$$

$$\theta_2 \leftarrow \theta_2 + \gamma \nabla_{\theta_2} \hat{J}$$

Output: Trained dual potential networks $\phi_{\theta_1}, \psi_{\theta_2}$

Algorithmic Implementations

When the reference process is a reversible Brownian motion with volatility $\sigma > 0$, the joint endpoint distribution $R_{01}(x, y)$ is governed by a Gaussian transition kernel. Assuming a uniform or Lebesgue prior on the initial state for the stationary reversible process, the joint density is proportional to the transition probability:

$$R_{01}(x, y) \propto \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

Taking the natural logarithm isolates the quadratic penalty:

$$\log R_{01}(x, y) = -\frac{1}{2\sigma^2}\|x - y\|^2 + C$$

where C is an additive normalizing constant that vanishes upon differentiation.

Because we have the exact analytical form of the log-prior, the intractable joint score $\nabla_{x,y} \log R_{01}(x, y)$ that previously required a neural network (s_ω) now resolves into two exact, linear vector fields:

$$\nabla_x \log R_{01}(x, y) = -\frac{1}{2\sigma^2}\nabla_x \left(\|x - y\|^2\right) = \frac{y - x}{\sigma^2}$$

$$\nabla_y \log R_{01}(x, y) = -\frac{1}{2\sigma^2}\nabla_y \left(\|x - y\|^2\right) = \frac{x - y}{\sigma^2}$$

Notice that these two score vectors are perfectly symmetric and actively pull the joint particles (x, y) toward each other, behaving as an elastic spring with stiffness proportional to $1/\sigma^2$.

Substituting these exact scores into the Unadjusted Langevin Algorithm (ULA), the discrete update rules for sampling from the joint optimal coupling $\pi_{\theta_1\theta_2}$ become entirely free of auxiliary neural networks. Given a step size $\eta > 0$ and standard Gaussian noise $z_x^{(k)}, z_y^{(k)} \sim \mathcal{N}(0, I)$, the coupled diffusion is:

$$x^{(k)} = x^{(k-1)} + \frac{\eta}{2}\text{stop_grad}\left(\nabla_x \phi_{\theta_1}(x^{(k-1)}) + \frac{y^{(k-1)} - x^{(k-1)}}{\sigma^2}\right) + \sqrt{\eta} z_x^{(k)}$$

$$y^{(k)} = y^{(k-1)} + \frac{\eta}{2}\text{stop_grad}\left(\nabla_y \psi_{\theta_2}(y^{(k-1)}) + \frac{x^{(k-1)} - y^{(k-1)}}{\sigma^2}\right) + \sqrt{\eta} z_y^{(k)}$$

Once the joint Langevin simulation converges to yield the generated batch $(\tilde{X}, \tilde{Y}) \sim \pi_{\theta_1\theta_2}$, the targets for the dual potentials remain the standard empirical Contrastive

Divergence gradients. The neural networks are optimized to maximize the potential of the real data and minimize the potential of the generated data:

$$\nabla_{\theta_1} \hat{J} = \mathbb{E}_{x \sim \mu}[\nabla_{\theta_1} \phi_{\theta_1}(x)] - \mathbb{E}_{\tilde{x} \sim \tilde{X}}[\nabla_{\theta_1} \phi_{\theta_1}(\tilde{x})]$$

$$\nabla_{\theta_2} \hat{J} = \mathbb{E}_{y \sim \nu}[\nabla_{\theta_2} \psi_{\theta_2}(y)] - \mathbb{E}_{\tilde{y} \sim \tilde{Y}}[\nabla_{\theta_2} \psi_{\theta_2}(\tilde{y})]$$